Build automated data quality & regression testing framework for Terminus platform (PCBDF-2899)

[PCBDF-3032] Spike: Investigate implementation options for automated data quality & regression testing framework for Terminus

Created: 13/Mar/26 Updated: 23/Mar/26

Status:	In Progress
Project:	PCB: Data Foundation
Components:	None
Affects versions:	None
Fix versions:	None
Parent:	Build automated data quality & regression testing framework for Terminus platform

Type:	Spike	**Priority:**	4-Low
Reporter:	Chris Chalissery	**Assignee:**	Harshit Patel
Resolution:	Unresolved	**Votes:**	0
Labels:	None		
Remaining Estimate:	Not Specified		
Time Spent:	Not Specified		
Original estimate:	Not Specified		

Epic Link:	Build automated data quality & regression testing framework for Terminus platform
Team:	
Sprint:	Centaurs-S2026-04

Description

As a Data engineer ,We need to investigate and propose a concrete implementation approach for the Terminus automated data quality & regression testing framework described in epic PCBDF-2899 TO DO (https://pc-technology.atlassian.net/browse/PCBDF-2899), aligned with the Terminus Data Platform Strategy (https://pc-technology.atlassian.net/wiki/spaces/PCFENG/pages/8610054409/Terminus+Data+Platform+Strategy) and the Terminus team's DoD / test strategy (https://pc-technology.atlassian.net/wiki/spaces/OT/pages/6261342209/Terminus+Data+Platform).

This spike should answer:

1. **Architecture & Tooling Options**
   - What tools / patterns should we use for:
     - Schema validation and data quality checks in BigQuery (e.g., dbt tests, Great Expectations, Dataplex quality scans, custom BQ-based checks)?
     - Regression testing across DAGs when:
       - shared Python libs or common DAG creators change
       - pipeline logic changes
       - Terraform/Terragrunt infrastructure changes
   - How can we leverage / extend:
     - Existing Dataplex quality scans and quality reports
     - Existing Airflow DAG patterns and common libraries
     - Existing observability stack (Slack alerts, Grafana, log-based metrics)?
2. **Test Declaration & Configuration**
   - Recommended standard for declaring tests (e.g., YAML configs in repo, BQ metadata, Dataplex rules).
   - How domain teams can:
     - Opt into platform baseline tests
     - Add/override domain-specific checks without breaking standardization.
   - How this fits with existing patterns for schema definition and table creation described in the Terminus strategy page.
3. **CI/CD Integration & Impact Analysis**
   - How to integrate the framework into current CI/CD:
     - Trigger tests on PR and on merge to `main/master` for:
       - DAG code
       - shared libraries
       - Terraform/Terragrunt changes

        - Run scheduled regression suites (e.g., nightly) for critical DAGs.
    - What options exist for **impact analysis**:
        - How to determine which pipelines / datasets must be tested when a shared library or infra changes, without running tests for all ~1,000+ DAGs every time.
    - How to provide clear pass/fail signals to CI to block releases on critical failures.
4. **Monitoring, Reporting & Support Model**
    - Options for platform-level dashboards:
        - Test coverage & status by domain, dataset, pipeline
        - Quality metrics (error rates, recurring rule failures).
    - How to integrate with:
        - Slack alerts
        - Existing Grafana dashboards.
    - Initial proposal for:
        - Framework ownership (which team)
        - Incident handling & escalation (Tier 1–3)
        - Integration with Jira for automated ticket creation on recurring failures.
5. **Data Handling**.
    - Using non-prod safe datasets (sampled / obfuscated) for regression tests.
    - RBAC model for running tests in BigQuery, Dataplex, etc.
6. **Onboarding, test engineers Experience & Governance**
    - What the test engineers experience should look like for outcome teams:
        - How they add tests to new / existing tables or DAGs.
    - How this framework ties into:
        - Team DoD / UAT strategy on Terminus page (https://pc-technology.atlassian.net/wiki/spaces/OT/pages/6261342209/Terminus+Data+Platform)
        - Future AI-powered regression testing vision from the strategy page.

Acceptance Criteria

- A **written design / spike outcome document** is created and linked in this ticket that includes:
    - Recommended high-level architecture for the data quality & regression testing framework.
    - Comparison of at least 2–3 tooling options (e.g., Dataplex vs external framework vs custom BQ-based solution) with pros/cons and fit for Terminus.
    - Proposed standard for test declaration (file format, location, examples).
    - CI/CD integration approach (how tests are triggered, where results surface, how failures block releases).
    - Conceptual design for impact analysis (how to select which pipelines/datasets to test after a change).
    - Recommended support model & ownership proposal.
- The document includes **at least one end-to-end example**:
    - Example DAG / dataset:
        - how tests are defined,
        - how they run in CI,
        - how results are reported and consumed by an outcome team.
- Open questions, assumptions, and risks are explicitly documented, including:
    - Dependencies on other initiatives (e.g., Dataplex rollout, schema standardization, BigQuery table standardization work).
    - Phased rollout approach (MVP vs future enhancements, including AI/automation capabilities).

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